



Structure of a URL

A URL is composed of several components, each serving a specific purpose:

- Scheme
 - Specifies the protocol used to access the resource
 - Example: HTTP, HTTPS, FTP, MAILTO...
 - Separated from the host by a colon and two slashes "://"
 - Format: "<scheme>://"
- Authentication (optional)
 - Includes credentials for accessing the resource
 - Separated from the rest of the URL by an at sign '@'
 - Format: "<user>:<password>@"
 - Example: "https://john.doe:secret123@example.com/dashboard"
- Host
 - Indicates the Domain Name or IP Address of the server where the resource is hosted
 - Example: "www.example.com", "192.0.2.1"
- Port (optional)
 - Specifies the Port Number on the server to connect to. It's omitted if the default port for the scheme is used
 - Example: ":80" for HTTP, ":443" for HTTPS
 - Separated from the host by a colon ':'

- Format: "http://example.com:<port>"
- Path
 - Provides the specific location of the resource on the server
 - Example: "/about", "/images/photo.jpg"
 - Separated from the host or port by a slash '/'
 - Format: "http://example.com:8080/<path>"
- Query (optional)
 - Contains Query Params used to provide additional information on the server, used for searching or filtering
 - Example: "?search=networking&sort=descending"
 - Separated from the path by a question mark '?'
 - Format: "http://example.com:8080/searching?<query>"
- Fragment (optional)
 - Refers to a specific section within the resource, such as an anchor within a webpage
 - Used to navigate to a specific section within the page
 - Example: "https://example.com/page#section2"
 - Separated from the rest of the URL by a number sign '#'
 - Format: "http://example.com:8080/searching#<fragment>"